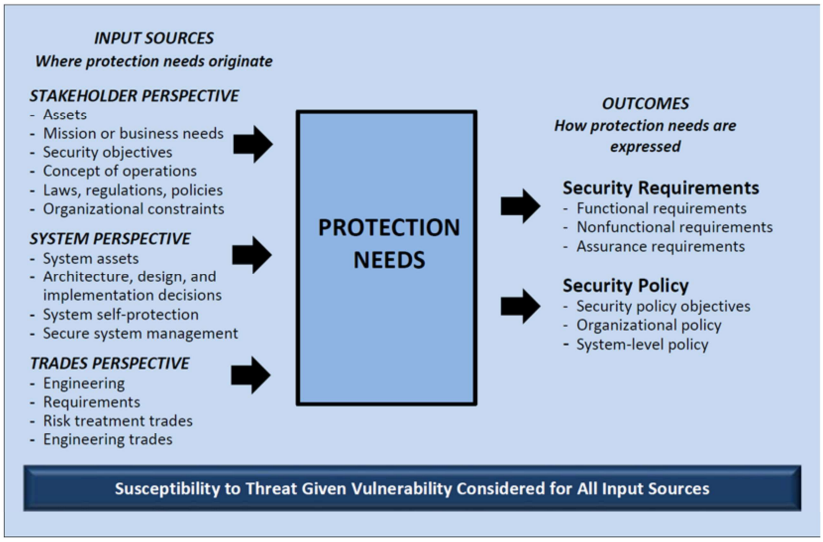
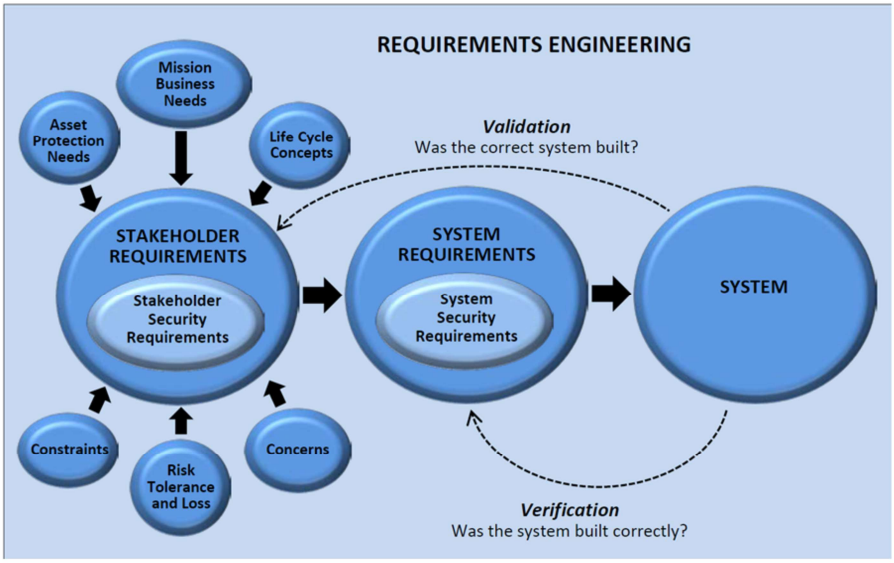
System Security Engineering Protection Needs



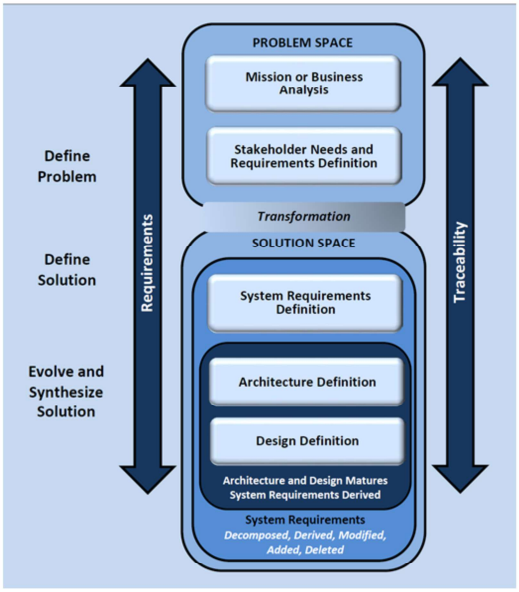
© 2018 Al-Nafi. All Rights Reserved.

Stakeholder and System Requirements



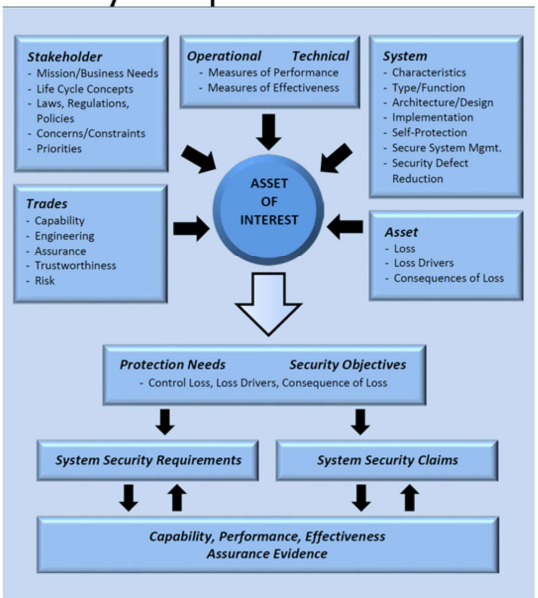
© 2018 Al-Nafi. All Rights Reserved.

Requirements Engineering across key life cycle processes



© 2018 Al-Nafi. All Rights Reserved.

Factors in security requirements analysis



© 2018 Al-Nafi. All Rights Reserved.

Security Policy focuses squarely on CIA

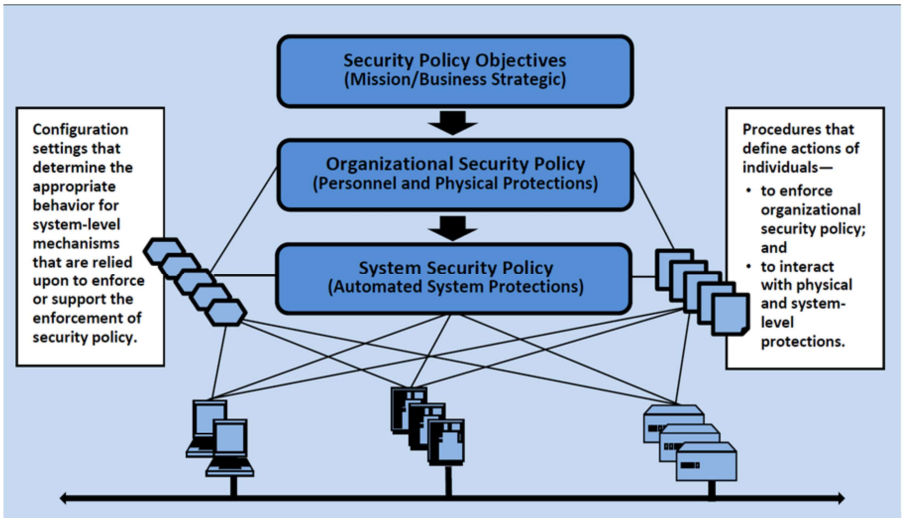
Confidentiality concerns are fully within the scope of security disciplines and therefore, security policy.



Integrity and availability concerns span security and non-security disciplines such as performance, fault tolerance, reliability, safety, and functional correctness. Security policy identifies the security-relevant aspects of integrity and availability.

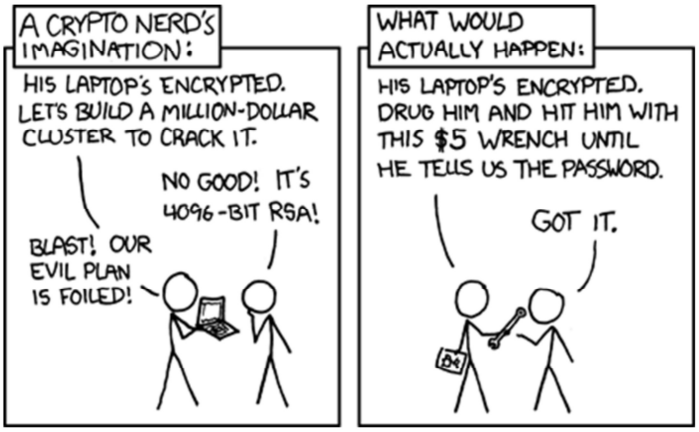
© 2018 Al-Nafi. All Rights Reserved.

Allocation of security Policy enforcement responsibilities



© 2018 Al-Nafi. All Rights Reserved.

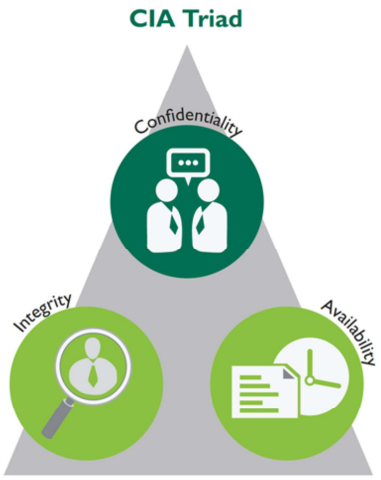
Information **Security Models**



© 2018 Al-Nafi. All Rights Reserved.

Security models define rules of behavior for an information system to enforce policies related to system security but typically involving confidentiality and/or integrity policies of the system. Models define allowable behavior for one or more aspect of system operation. When implemented in a system, technology enforces the rules of behavior to ensure security goals (e.g., confidentiality, integrity) are met.

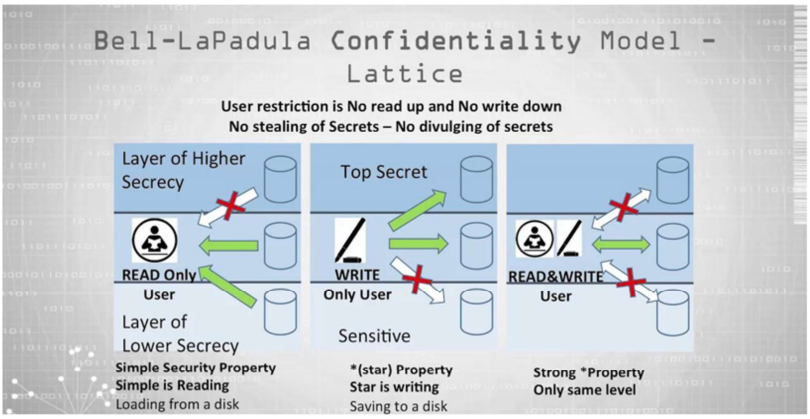
Security Models foundation



© 2018 Al-Nafi. All Rights Reserved.

A key principle of Systems Security Engineering and a differentiator from traditional Systems Engineering is that Systems Security Engineering is focused on supporting the confidentiality, integrity, and availability (CIA) needs of the system and not on the system functional requirements. This is known as the CIA triad and is a prime governing factor for all system security engineering activities.

Bell-LaPadula Model



© 2018 Al-Nafi. All Rights Reserved.

The Bell–LaPadula (BLP) model is intended to address confidentiality in a multilevel security (MLS) system. It defines two primary security constructs, subjects and objects. Subjects are the active parties, while objects are the passive parties. To help determine what subjects will be allowed to do, they are assigned clearances that outline what modes of access (e.g., read, write) they will be allowed to use when
they interact with objects.

The model system uses labels to keep track of clearances and classifications and implements a set of rules to limit interactions between different types of subjects and objects. It was an early security model and does not provide a mechanism for a one-tone mapping of individual subjects and objects. This also needs
to be addressed by other models or features within a practical operating system.

The model defines two properties, the ss-property and the *-property.

Simple Security property: A subject cannot read/access an object of a higher classification (no read up)

Star property: A subject can only save an object at the same or higher classification (no write down)

The model does not attempt to define technical constructs or solutions. It merely identifies a high level set of rules that if implemented correctly, prevent the exposure or unauthorized disclosure of information in a system processing different classification levels of data.